GIK INSTITUTE
OF ENGINEERING SCIENCES AND TECHNOLOGY

2025/02

Student Name:	Abdullah Ejaz Janjua
Reg No:	2023038
Assignment No:	01
Instructor:	Dr. Khurram Khan Jadoon

FACULTY OF COMPUTER SCIENCE AND ENGINEERING

Ghulam Ishaq Institue of Engineering Sciences and Technology, Topi, Swabi

Question 01

1. **Suppose the agent wants to take the leftmost path from the start (0,0) facing north to the goal (0,4) facing west. Write down the sequence of actions for it to take.**

   - Turn Right
   - Move Fast (v = 1)
   - Move Slow (v = 0)
   - Turn Left
   - Move Fast (v = 1)
   - Move maintain (v = 1)
   - Move slow (v = 0)
   - Turn Left
   - Move Fast (v = 1)
   - Move Slow (v = 0)
   - Turn Right
   - Move Fast (v = 1)
   - Move Maintain (v = 1)
   - Move slow (v = 0)
   - Turn Left

2. **If the grid is M by N, what is the size of the state space? You should assume that all configurations are reachable from the start state.**

   We have MxN total positions, 4 directions to face, and we can have velocity from $0 \leq v \leq v_{max}$. Thus, our state space will be:

   $$state\_space = M * N * 4 * (v_{max} + 1)$$

3. **What is the maximum branching factor of this problem? Draw an example state (x, y, orientation, velocity, grid/walls) that has this branching factor, and list the set of available actions.**

   To achieve maximum branching factor, we need to in a state where we can move anywhere. Consider the state ((0, 3), N, 0), we have following legal actions that will result in a new state:

   (a) Turn Left
   (b) Turn Right
   (c) Move Fast

   This would give us a **maximum branching factor of 3**. We do not consider the action Move Maintain as that would not result in a new state. Refer to Figure 1 for an example state that shows this branching factor.

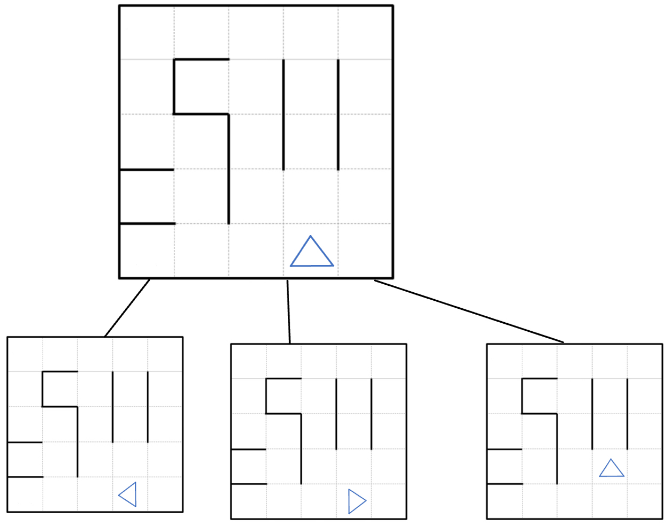

Figure 1: An example state that has maximum branching factor

4. **What is the minimum branching factor? Draw an example state (x, y, orientation, velocity, grid/walls) that has this branching factor, and list the set of available actions.**

   Consider the state ((0, 4), E, $v > 1$), at this state there are no legal moves available to us, giving us a minimum branching factor of 0.
   Refer to Figure 2 for an example state.

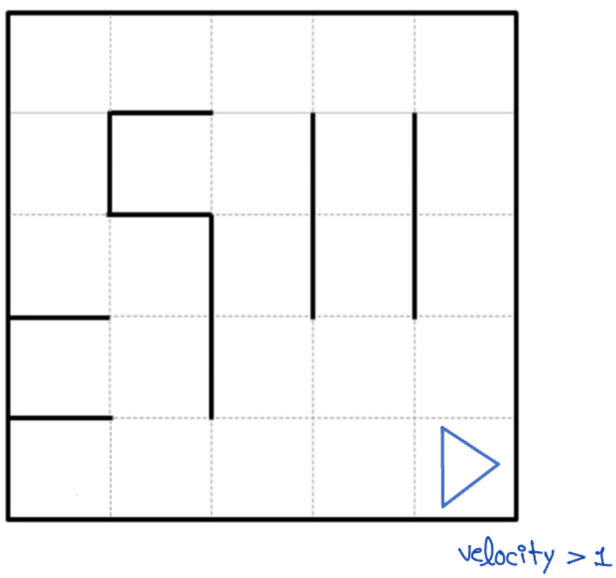

Figure 2: An example state that has minimum branching factor

Question 02

1. **State Representation: Describe how you would represent the state of the system at any given moment.**

   Suppose we have R rings that we want to move. To do that we will need three locations: 1) Initial position (where rings will spawn in), 2) Goal position (where we want to move the rings to.), 3) Intermediate position (to store rings in a temporary location). So, this gives us a $R \times 3$ grid.
   Our state at any given moment will be represented as:

   $$\text{(sorted\_rings\_at\_initial, sorted\_rings\_at\_intermediate, sorted\_rings\_at\_goal).}$$

   Each sorted_rings_at_position is a sorted list $[R \dots 1]$. Note that each element in the list corresponds to a ring's size i.e R is size of the largest ring and $R - 1$ is the size of second largest ring and so on.

2. **State Space Size: What is the size of the state space for this problem (considering any number of rings)?**

   At each location, there can be at most R rings and we have 3 locations. So, we have:

   $$state\_size = 3^R$$

   As our state space excludes the possibility of illegal states by ensuring that each position has rings sorted in decreasing order i.e largest on the bottom.

3. **Start State: How would you represent the initial state when all rings are stacked at the landing site?**

   Our initial state will be as:

   $$([R \dots 1], 0, 0)$$

   Where R is our number of rings.

4. **Legal Actions: Given a state, what are the possible actions you can take (moving rings)? How would you represent these actions?**

   Our actions involve moving rings from one position to another. However, moving a larger ring on top of a smaller ring will be considered an illegal action.

5. **Goal Test: How would you determine if a state represents the goal state where all rings are at the research station in the correct order?**

   If all of our rings at the goal postion in sorted order and other positios are empty then that will be our goal state. Our goal state will be as:

   $$(0, 0, [R \dots 1])$$

   Where R is our number of rings.

Question 03

Please refer to the attached file code.py